

Module 3: Correlation and Regression

Expectation of Random variable:

Mathematical Expectation:

Discrete case:

Let x be a discrete random variable with probability density function $P(x)$. Then the mathematical expectation of X is denoted by $E(X)$ and is given by

$$E(X) = \sum_x xP(x)$$

Continuous case:

Let X be a continuous random variable with probability density function $f(x)$. Then the mathematical expectation of X is

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

Moments: r^{th} moment about mean:

Let X be a random variable and $E(X) = \mu$ be the mean then the r^{th} moment about mean is denoted by μ_r and defined as

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$$\mu_r = E(X - E(X))^r = \int_{-\infty}^{\infty} (x - E(x))^r f(x) dx = \int_{-\infty}^{\infty} (x - \bar{x})^r f(x) dx, \quad \text{where } E(X) = \mu = \bar{x}$$

$$\text{where } r = 1, \mu_1 = \int_{-\infty}^{\infty} (x - \bar{x}) f(x) dx = \int_{-\infty}^{\infty} x f(x) dx - \int_{-\infty}^{\infty} \bar{x} f(x) dx = \bar{x} - \bar{x} \int_{-\infty}^{\infty} f(x) dx$$

$$= \bar{x} - \bar{x} = 0 \quad \left\{ \because \int_{-\infty}^{\infty} f(x) dx = 1 \right\}$$

$$\mu_1 = 0$$

When $r = 2$

$$\mu_2 = \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x) dx, \text{ which gives the variance.}$$

Theorems on Expectation:

1. Expectation of constant is constant. $E(a) = a$.
2. If X and Y are independent. Then $E(XY) = E(X) \cdot E(Y)$

Note:

$$1. E[a + G(x)] = a + E[G(x)]$$

$$2. E\left(\frac{1}{X}\right) \neq \frac{1}{E(X)}$$

$$3. E(\log X) \neq \log E(X)$$

$$4. E(X^2) \neq [E(X)]^2$$

Variance of a Random variable:

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The variance of a random variable X which is represented as $V(x)$ is defined as the expectation of squares of the deviations from the expected value

$$\text{i.e., } V(X) = E[X - E(X)]^2 = E(X^2) - [E(X)]^2$$

Theorems:

1. Variance of a constant is Zero.

$$V(a) = E[a - E(a)]^2 = E[a - a]^2 = 0$$

2. $V(aX) = a^2V(X)$, where a is a constant.

Co-Variance:

Let X and Y be random variables then, covariance between X and Y is denoted by $COV(X, Y)$ and defined as,

$$\begin{aligned} \text{cov}(X, Y) &= E\{X - E(X)(Y - E(Y))\} \\ &= E\{XY - XE(Y) - YE(X) + E(X)E(Y)\} \\ &= E(XY) - E(Y)E(X) - E(X)E(Y) + E(X)E(Y) \\ &= E(XY) - E(Y)E(X) \end{aligned}$$

$$COV(X, Y) = E(XY) - E(Y)E(X)$$

Note:

1. If X and Y are independent $E(XY) = E(X).E(Y)$

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$$\begin{aligned} \text{COV}(X, Y) &= E(XY) - E(Y)E(X) \\ &= E(X) \cdot E(Y) - E(Y)E(X) = 0 \end{aligned}$$

2. If X is a random variable, then $V(ax+b)=a^2V(x)$, where a and b are constants.

Problem:

1. Given the following probability distribution of X.

Find (1) $E(X)$ (2) $E(X^2)$ (3) $E(2X \pm 3)$ (4) $\text{Var}(2X \pm 3)$

X	-3	-2	-1	0	1	2	3
P(X)	0.05	0.10	0.30	0	0.30	0.15	0.10

$$\begin{aligned} 1) E(X) &= \sum XP(X) \\ &= (-3)(0.05) + (-2)(0.10) + (-1)(0.30) + 0 + (1)(0.30) + (2)(0.15) + (3)(0.10) \\ &= 0.25 \end{aligned}$$

$$\begin{aligned} 2) \\ E(X^2) &= (-3)^2(0.05) + \dots + (3)^2(0.10) = 2.95 \end{aligned}$$

$$\begin{aligned} 3) \\ E(2X \pm 3) &= 2E(X) \pm 3 \\ &= 2(0.25) \pm 3 \\ &= 0.5 \pm 3 \end{aligned}$$

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4)

$$\text{var}(2X \pm 3) = 2^2 \text{var}(X)$$

$$\text{But } \text{var}(X) = E(X^2) - E(X)^2 = 2.8875$$

$$\therefore \text{Var}(2X \pm 3) = 4(2.8875) = 11.55$$

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